

5 IDENTICAL PARTICLES

5.1 TWO-PARTICLE SYSTEMS

For a *single* particle, $\Psi(\mathbf{r}, t)$ is a function of the spatial coordinates, $\mathbf{r}$, and the time, t (I'll ignore spin, for the moment). The state of a *two*-particle system is a function of the coordinates of particle one ($\mathbf{r}_1$), the coordinates of particle two ($\mathbf{r}_2$), and the time:

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, t). \quad (5.1)$$

Its time evolution is determined by the Schrödinger equation:

$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi, \quad (5.2)$$

where H is the Hamiltonian for the whole works:

$$\hat{H} = -\frac{\hbar^2}{2m_1} \nabla_1^2 - \frac{\hbar^2}{2m_2} \nabla_2^2 + V(\mathbf{r}_1, \mathbf{r}_2, t) \quad (5.3)$$

(the subscript on ∇ indicates differentiation with respect to the coordinates of particle 1 or particle 2, as the case may be). The statistical interpretation carries over in the obvious way:

$$|\Psi(\mathbf{r}_1, \mathbf{r}_2, t)|^2 d^3\mathbf{r}_1 d^3\mathbf{r}_2 \quad (5.4)$$

is the probability of finding particle 1 in the volume $d^3\mathbf{r}_1$ and particle 2 in the volume $d^3\mathbf{r}_2$; as always, Ψ must be normalized:

$$\int |\Psi(\mathbf{r}_1, \mathbf{r}_2, t)|^2 d^3\mathbf{r}_1 d^3\mathbf{r}_2 = 1. \quad (5.5)$$

For time-independent potentials, we obtain a complete set of solutions by separation of variables:

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, t) = \psi(\mathbf{r}_1, \mathbf{r}_2) e^{-iEt/\hbar}, \quad (5.6)$$

where the spatial wave function (ψ) satisfies the time-independent Schrödinger equation:

$$-\frac{\hbar^2}{2m_1} \nabla_1^2 \psi - \frac{\hbar^2}{2m_2} \nabla_2^2 \psi + V\psi = E\psi, \quad (5.7)$$

and E is the total energy of the system. In general, solving Equation 5.7 is difficult, but two special cases can be reduced to one-particle problems:

1. Noninteracting particles. Suppose the particles do not interact with one another, but each is subject to some external force. For example, they might be attached to two different springs. In that case the total potential energy is the *sum* of the two:

$$V(\mathbf{r}_1, \mathbf{r}_2) = V_1(\mathbf{r}_1) + V_2(\mathbf{r}_2), \quad (5.8)$$

and Equation 5.7 can be solved by separation of variables:

$$\psi(\mathbf{r}_1, \mathbf{r}_2) = \psi_a(\mathbf{r}_1)\psi_b(\mathbf{r}_2). \quad (5.9)$$

Plugging Equation 5.9 into Equation 5.7, dividing by $\psi(\mathbf{r}_1, \mathbf{r}_2)$, and collecting the terms in $\mathbf{r}_1$ alone and in $\mathbf{r}_2$ alone, we find that $\psi_a(\mathbf{r}_1)$ and $\psi_b(\mathbf{r}_2)$ each satisfy the one-particle Schrödinger equation:

$$\begin{aligned} -\frac{\hbar^2}{2m_1}\nabla_1^2\psi_a(\mathbf{r}_1) + V_1(\mathbf{r}_1)\psi_a(\mathbf{r}_1) &= E_a\psi_a(\mathbf{r}_1), \\ -\frac{\hbar^2}{2m_2}\nabla_2^2\psi_b(\mathbf{r}_2) + V_2(\mathbf{r}_2)\psi_b(\mathbf{r}_2) &= E_b\psi_b(\mathbf{r}_2), \end{aligned} \quad (5.10)$$

and $E = E_a + E_b$. In this case the two-particle wave function is a simple *product* of one-particle wave functions,

$$\begin{aligned} \Psi(\mathbf{r}_1, \mathbf{r}_2, t) &= \psi_a(\mathbf{r}_1)\psi_b(\mathbf{r}_2)e^{-i(E_a+E_b)t/\hbar} \\ &= \left(\psi_a(\mathbf{r}_1)e^{-iE_at/\hbar}\right)\left(\psi_b(\mathbf{r}_2)e^{-iE_bt/\hbar}\right) = \Psi_a(\mathbf{r}_1, t)\Psi_b(\mathbf{r}_2, t), \end{aligned} \quad (5.11)$$

and it makes sense to say that particle 1 is in state a , and particle 2 is in state b . But any linear combination of such solutions will still satisfy the (time-dependent) Schrödinger equation—for instance

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, t) = \frac{3}{5}\Psi_a(\mathbf{r}_1, t)\Psi_b(\mathbf{r}_2, t) + \frac{4}{5}\Psi_c(\mathbf{r}_1, t)\Psi_d(\mathbf{r}_2, t). \quad (5.12)$$

In this case the state of particle 1 depends on the state of particle 2, and vice versa. If you measured the energy of particle 1, you might get E_a (with probability 9/25), in which case the energy of particle 2 is definitely E_b , or you might get E_c (probability 16/25), in which case the energy of particle 2 is E_d . We say that the two particles are **entangled** (Schrödinger's lovely term). An entangled state is one that *cannot* be written as a product of single-particle states.¹

2. Central potentials. Suppose the particles interact *only* with one another, via a potential that depends on their separation:

$$V(\mathbf{r}_1, \mathbf{r}_2) \rightarrow V(|\mathbf{r}_1 - \mathbf{r}_2|). \quad (5.13)$$

The hydrogen atom would be an example, if you include the motion of the proton. In this case the two-body problem reduces to an equivalent one-body problem, just as it does in *classical* mechanics (see Problem 5.1).

In general, though, the two particles will be subject both to external forces *and* to mutual interactions, and this makes the analysis more complicated. For example, think of the two electrons in a helium atom: each feels the Coulomb attraction of the nucleus (charge $2e$), and at the same time they repel one another:

$$V(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{4\pi\epsilon_0} \left(-\frac{2e^2}{|\mathbf{r}_1|} - \frac{2e^2}{|\mathbf{r}_2|} + \frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|} \right). \quad (5.14)$$

We'll take up this problem in later sections.

¹ The classic example of an entangled state is two spin-1/2 particles in the singlet configuration (Equation 4.176).

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Problem 5.1 A typical interaction potential depends only on the vector displacement $\mathbf{r} \equiv \mathbf{r}_1 - \mathbf{r}_2$ between the two particles: $V(\mathbf{r}_1, \mathbf{r}_2) \rightarrow V(\mathbf{r})$. In that case the Schrödinger equation separates, if we change variables from $\mathbf{r}_1, \mathbf{r}_2$ to $\mathbf{r}$ and $\mathbf{R} \equiv (m_1\mathbf{r}_1 + m_2\mathbf{r}_2) / (m_1 + m_2)$ (the center of mass).

- (a) Show that $\mathbf{r}_1 = \mathbf{R} + (\mu/m_1)\mathbf{r}$, $\mathbf{r}_2 = \mathbf{R} - (\mu/m_2)\mathbf{r}$, and $\nabla_1 = (\mu/m_2)\nabla_R + \nabla_r$, $\nabla_2 = (\mu/m_1)\nabla_R - \nabla_r$, where

$$\mu \equiv \frac{m_1 m_2}{m_1 + m_2} \quad (5.15)$$

is the **reduced mass** of the system.

- (b) Show that the (time-independent) Schrödinger equation (5.7) becomes

$$-\frac{\hbar^2}{2(m_1 + m_2)}\nabla_R^2\psi - \frac{\hbar^2}{2\mu}\nabla_r^2\psi + V(\mathbf{r})\psi = E\psi.$$

- (c) Separate the variables, letting $\psi(\mathbf{R}, \mathbf{r}) = \psi_R(\mathbf{R})\psi_r(\mathbf{r})$. Note that ψ_R satisfies the one-particle Schrödinger equation, with the *total* mass $(m_1 + m_2)$ in place of m , potential zero, and energy E_R , while ψ_r satisfies the one-particle Schrödinger equation with the *reduced* mass in place of m , potential $V(\mathbf{r})$, and energy E_r . The total energy is the sum: $E = E_R + E_r$. What this tells us is that the center of mass moves like a free particle, and the *relative* motion (that is, the motion of particle 2 with respect to particle 1) is the same as if we had a *single* particle with the *reduced* mass, subject to the potential V . Exactly the same decomposition occurs in *classical* mechanics;² it reduces the two-body problem to an equivalent one-body problem.

Problem 5.2 In view of Problem 5.1, we can correct for the motion of the nucleus in hydrogen by simply replacing the electron mass with the reduced mass.

- (a) Find (to two significant digits) the percent error in the binding energy of hydrogen (Equation 4.77) introduced by our use of m instead of μ .
- (b) Find the separation in wavelength between the red Balmer lines ($n = 3 \rightarrow n = 2$) for hydrogen and deuterium (whose nucleus contains a neutron as well as the proton).
- (c) Find the binding energy of **positronium** (in which the proton is replaced by a positron—positrons have the same mass as electrons, but opposite charge).
- (d) Suppose you wanted to confirm the existence of **muonic hydrogen**, in which the electron is replaced by a muon (same charge, but 206.77 times heavier). Where (i.e. at what wavelength) would you look for the “Lyman- α ” line ($n = 2 \rightarrow n = 1$)?

² See, for example, Jerry B. Marion and Stephen T. Thornton, *Classical Dynamics of Particles and Systems*, 4th edn, Saunders, Fort Worth, TX (1995), Section 8.2.

Problem 5.3 Chlorine has two naturally occurring isotopes, Cl^{35} and Cl^{37} . Show that the vibrational spectrum of HCl should consist of closely spaced doublets, with a splitting given by $\Delta\nu = 7.51 \times 10^{-4}\nu$, where ν is the frequency of the emitted photon. *Hint:* Think of it as a harmonic oscillator, with $\omega = \sqrt{k/\mu}$, where μ is the reduced mass (Equation 5.15) and k is presumably the same for both isotopes.

5.1.1 Bosons and Fermions

Suppose we have two noninteracting particles, number 1 in the (one-particle) state $\psi_a(\mathbf{r})$, and number 2 in the state $\psi_b(\mathbf{r})$. In that case $\psi(\mathbf{r}_1, \mathbf{r}_2)$ is the product (Equation 5.9):

$$\psi(\mathbf{r}_1, \mathbf{r}_2) = \psi_a(\mathbf{r}_1)\psi_b(\mathbf{r}_2). \quad (5.16)$$

Of course, this assumes that we can tell the particles apart—otherwise it wouldn't make any sense to claim that number 1 is in state ψ_a and number 2 is in state ψ_b ; all we could say is that *one* of them is in the state ψ_a and the other is in state ψ_b , but we wouldn't know which is which. If we were talking *classical* mechanics this would be a silly objection: You can *always* tell the particles apart, in principle—just paint one of them red and the other one blue, or stamp identification numbers on them, or hire private detectives to follow them around. But in quantum mechanics the situation is fundamentally different: You can't paint an electron red, or pin a label on it, and a detective's observations will inevitably and unpredictably alter its state, raising the possibility that the two particles might have secretly switched places. The fact is, all electrons are *utterly identical*, in a way that no two classical objects can ever be. It's not just that *we* don't know which electron is which; *God* doesn't know which is which, because there is really *no such thing* as “this” electron, or “that” electron; all we can legitimately speak about is “an” electron.

Quantum mechanics neatly accommodates the existence of particles that are *indistinguishable in principle*: We simply construct a wave function that is *noncommittal* as to which particle is in which state. There are actually two ways to do it:

$$\psi_{\pm}(\mathbf{r}_1, \mathbf{r}_2) = A [\psi_a(\mathbf{r}_1)\psi_b(\mathbf{r}_2) \pm \psi_b(\mathbf{r}_1)\psi_a(\mathbf{r}_2)]; \quad (5.17)$$

the theory admits two kinds of identical particles: **bosons** (the plus sign), and **fermions** (the minus sign). Boson states are **symmetric** under interchange, $\psi_+(\mathbf{r}_2, \mathbf{r}_1) = \psi_+(\mathbf{r}_1, \mathbf{r}_2)$; fermion states are **antisymmetric** under interchange, $\psi_-(\mathbf{r}_2, \mathbf{r}_1) = -\psi_-(\mathbf{r}_1, \mathbf{r}_2)$. It so happens that

$$\left\{ \begin{array}{l} \text{all particles with integer spin are bosons, and} \\ \text{all particles with half integer spin are fermions.} \end{array} \right. \quad (5.18)$$

This **connection between spin and statistics** (bosons and fermions have quite different statistical properties) can be *proved* in *relativistic* quantum mechanics; in the nonrelativistic theory it is simply taken as an axiom.³

³ It seems strange that *relativity* should have anything to do with it, and there has been a lot of discussion as to whether it might be possible to prove the spin-statistics connection in other ways. See, for example, Robert C. Hilborn, *Am. J. Phys.* **63**, 298 (1995); Ian Duck and E. C. G. Sudarshan, *Pauli and the Spin-Statistics Theorem*, World Scientific, Singapore (1997). For a comprehensive bibliography on spin and statistics see C. Curceanu, J. D. Gillaspay, and R. C. Hilborn, *Am. J. Phys.* **80**, 561 (2010).

It follows, in particular, that *two identical fermions* (for example, two electrons) *cannot occupy the same state*. For if $\psi_a = \psi_b$, then

$$\psi_-(\mathbf{r}_1, \mathbf{r}_2) = A [\psi_a(\mathbf{r}_1)\psi_a(\mathbf{r}_2) - \psi_a(\mathbf{r}_1)\psi_a(\mathbf{r}_2)] = 0,$$

and we are left with no wave function at all.⁴ This is the famous **Pauli exclusion principle**. It is not (as you may have been led to believe) a weird ad hoc assumption applying only to electrons, but rather a consequence of the rules for constructing two-particle wave functions, applying to *all* identical fermions.

Example 5.1

Suppose we have two noninteracting (they pass right through one another... never mind how you would set this up in practice!) particles, both of mass m , in the infinite square well (Section 2.2). The one-particle states are

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right), \quad E_n = n^2 K$$

(where $K \equiv \pi^2 \hbar^2 / 2ma^2$). If the particles are *distinguishable*, with number 1 in state n_1 and number 2 in state n_2 , the composite wave function is a simple product:

$$\psi_{n_1 n_2}(x_1, x_2) = \psi_{n_1}(x_1)\psi_{n_2}(x_2), \quad E_{n_1 n_2} = (n_1^2 + n_2^2) K.$$

For example, the ground state is

$$\psi_{11} = \frac{2}{a} \sin\left(\frac{\pi x_1}{a}\right) \sin\left(\frac{\pi x_2}{a}\right), \quad E_{11} = 2K;$$

the first excited state is doubly degenerate:

$$\psi_{12} = \frac{2}{a} \sin\left(\frac{\pi x_1}{a}\right) \sin\left(\frac{2\pi x_2}{a}\right), \quad E_{12} = 5K,$$

$$\psi_{21} = \frac{2}{a} \sin\left(\frac{2\pi x_1}{a}\right) \sin\left(\frac{\pi x_2}{a}\right), \quad E_{21} = 5K;$$

and so on. If the two particles are identical *bosons*, the ground state is unchanged, but the first excited state is *nondegenerate*:

$$\frac{\sqrt{2}}{a} \left[\sin\left(\frac{\pi x_1}{a}\right) \sin\left(\frac{2\pi x_2}{a}\right) + \sin\left(\frac{2\pi x_1}{a}\right) \sin\left(\frac{\pi x_2}{a}\right) \right]$$

(still with energy $5K$). And if the particles are identical *fermions*, there is *no* state with energy $2K$; the ground state is

$$\frac{\sqrt{2}}{a} \left[\sin\left(\frac{\pi x_1}{a}\right) \sin\left(\frac{2\pi x_2}{a}\right) - \sin\left(\frac{2\pi x_1}{a}\right) \sin\left(\frac{\pi x_2}{a}\right) \right],$$

and its energy is $5K$.

⁴ I'm still leaving out the spin, don't forget—if this bothers you (after all, a spinless fermion is an oxymoron), assume they're in the *same* spin state. I'll show you how spin affects the story in Section 5.1.3

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Problem 5.4

- (a) If ψ_a and ψ_b are orthogonal, and both are normalized, what is the constant A in Equation 5.17?
- (b) If $\psi_a = \psi_b$ (and it is normalized), what is A ? (This case, of course, occurs only for bosons.)

Problem 5.5

- (a) Write down the Hamiltonian for two noninteracting identical particles in the infinite square well. Verify that the fermion ground state given in Example 5.1 is an eigenfunction of $\hat{H}$, with the appropriate eigenvalue.
- (b) Find the next two excited states (beyond the ones given in the example)—wave functions, energies, and degeneracies—for each of the three cases (distinguishable, identical bosons, identical fermions).

5.1.2 Exchange Forces

To give you some sense of what the symmetrization requirement (Equation 5.17) actually *does*, I'm going to work out a simple one-dimensional example. Suppose one particle is in state $\psi_a(x)$, and the other is in state $\psi_b(x)$, and these two states are orthogonal and normalized. If the two particles are distinguishable, and number 1 is the one in state ψ_a , then the combined wave function is

$$\psi(x_1, x_2) = \psi_a(x_1)\psi_b(x_2); \quad (5.19)$$

if they are identical bosons, the composite wave function is (see Problem 5.4 for the normalization)

$$\psi_+(x_1, x_2) = \frac{1}{\sqrt{2}} [\psi_a(x_1)\psi_b(x_2) + \psi_b(x_1)\psi_a(x_2)]; \quad (5.20)$$

and if they are identical fermions, it is

$$\psi_-(x_1, x_2) = \frac{1}{\sqrt{2}} [\psi_a(x_1)\psi_b(x_2) - \psi_b(x_1)\psi_a(x_2)]. \quad (5.21)$$

Let's calculate the expectation value of the square of the separation distance between the two particles,

$$\langle (x_1 - x_2)^2 \rangle = \langle x_1^2 \rangle + \langle x_2^2 \rangle - 2\langle x_1 x_2 \rangle. \quad (5.22)$$

Case 1: Distinguishable particles. For the wave function in Equation 5.19,

$$\langle x_1^2 \rangle = \int x_1^2 |\psi_a(x_1)|^2 dx_1 \int |\psi_b(x_2)|^2 dx_2 = \langle x^2 \rangle_a$$

(the expectation value of x^2 in the one-particle state ψ_a),

$$\langle x_2^2 \rangle = \int |\psi_a(x_1)|^2 dx_1 \int x_2^2 |\psi_b(x_2)|^2 dx_2 = \langle x^2 \rangle_b,$$

and

$$\langle x_1 x_2 \rangle = \int x_1 |\psi_a(x_1)|^2 dx_1 \int x_2 |\psi_b(x_2)|^2 dx_2 = \langle x \rangle_a \langle x \rangle_b.$$

In this case, then,

$$\left\langle (x_1 - x_2)^2 \right\rangle_d = \langle x^2 \rangle_a + \langle x^2 \rangle_b - 2 \langle x \rangle_a \langle x \rangle_b. \quad (5.23)$$

(Incidentally, the answer would—of course—be the same if particle 1 had been in state ψ_b , and particle 2 in state ψ_a .)

Case 2: Identical particles. For the wave functions in Equations 5.20 and 5.21,

$$\begin{aligned} \langle x_1^2 \rangle &= \frac{1}{2} \left[\int x_1^2 |\psi_a(x_1)|^2 dx_1 \int |\psi_b(x_2)|^2 dx_2 \right. \\ &\quad + \int x_1^2 |\psi_b(x_1)|^2 dx_1 \int |\psi_a(x_2)|^2 dx_2 \\ &\quad \pm \int x_1^2 \psi_a(x_1)^* \psi_b(x_1) dx_1 \int \psi_b(x_2)^* \psi_a(x_2) dx_2 \\ &\quad \left. \pm \int x_1^2 \psi_b(x_1)^* \psi_a(x_1) dx_1 \int \psi_a(x_2)^* \psi_b(x_2) dx_2 \right] \\ &= \frac{1}{2} [\langle x^2 \rangle_a + \langle x^2 \rangle_b \pm 0 \pm 0] = \frac{1}{2} (\langle x^2 \rangle_a + \langle x^2 \rangle_b). \end{aligned}$$

Similarly,

$$\langle x_2^2 \rangle = \frac{1}{2} (\langle x^2 \rangle_b + \langle x^2 \rangle_a).$$

(Naturally, $\langle x_2^2 \rangle = \langle x_1^2 \rangle$, since you can't tell them apart.) But

$$\begin{aligned} \langle x_1 x_2 \rangle &= \frac{1}{2} \left[\int x_1 |\psi_a(x_1)|^2 dx_1 \int x_2 |\psi_b(x_2)|^2 dx_2 \right. \\ &\quad + \int x_1 |\psi_b(x_1)|^2 dx_1 \int x_2 |\psi_a(x_2)|^2 dx_2 \\ &\quad \pm \int x_1 \psi_a(x_1)^* \psi_b(x_1) dx_1 \int x_2 \psi_b(x_2)^* \psi_a(x_2) dx_2 \\ &\quad \left. \pm \int x_1 \psi_b(x_1)^* \psi_a(x_1) dx_1 \int x_2 \psi_a(x_2)^* \psi_b(x_2) dx_2 \right] \\ &= \frac{1}{2} (\langle x \rangle_a \langle x \rangle_b + \langle x \rangle_b \langle x \rangle_a \pm \langle x \rangle_{ab} \langle x \rangle_{ba} \pm \langle x \rangle_{ba} \langle x \rangle_{ab}) \\ &= \langle x \rangle_a \langle x \rangle_b \pm |\langle x \rangle_{ab}|^2, \end{aligned}$$

where

$$\langle x \rangle_{ab} \equiv \int x \psi_a(x)^* \psi_b(x) dx. \quad (5.24)$$

Thus

$$\left\langle (x_1 - x_2)^2 \right\rangle_{\pm} = \langle x^2 \rangle_a + \langle x^2 \rangle_b - 2 \langle x \rangle_a \langle x \rangle_b \mp 2 |\langle x \rangle_{ab}|^2. \quad (5.25)$$

Comparing Equations 5.23 and 5.25, we see that the difference resides in the final term:

$$\left\langle (\Delta x)^2 \right\rangle_{\pm} = \left\langle (\Delta x)^2 \right\rangle_d \mp 2 |\langle x \rangle_{ab}|^2; \quad (5.26)$$

identical bosons (the upper signs) tend to be somewhat closer together, and identical fermions (the lower signs) somewhat farther apart, than distinguishable particles in the same two states. Notice that $\langle x \rangle_{ab}$ *vanishes* unless the two wave functions actually *overlap*: if $\psi_a(x)$ is zero wherever $\psi_b(x)$ is *nonzero*, the integral in Equation 5.24 is zero. So if ψ_a represents an electron

in an atom in Chicago, and ψ_b represents an electron in an atom in Seattle, it's not going to make any difference whether you antisymmetrize the wave function or not. As a *practical* matter, therefore, it's okay to pretend that electrons with non-overlapping wave functions are distinguishable. (Indeed, this is the only thing that allows chemists to proceed at *all*, for in *principle* every electron in the universe is linked to every other one, via the antisymmetrization of their wave functions, and if this really *mattered*, you wouldn't be able to talk about any *one* unless you were prepared to deal with them *all*!)

The *interesting* case is when the overlap integral (Equation 5.24) is *not* zero. The system behaves as though there were a “force of attraction” between identical bosons, pulling them closer together, and a “force of repulsion” between identical fermions, pushing them apart (remember that we are for the moment ignoring spin). We call it an **exchange force**, although it's not really a force at all⁵—no physical agency is pushing on the particles; rather, it is a purely *geometrical* consequence of the symmetrization requirement. It is also a strictly quantum mechanical phenomenon, with no classical counterpart.

- * **Problem 5.6** Imagine two noninteracting particles, each of mass m , in the infinite square well. If one is in the state ψ_n (Equation 2.28), and the other in state ψ_l ($l \neq n$), calculate $\langle (x_1 - x_2)^2 \rangle$, assuming (a) they are distinguishable particles, (b) they are identical bosons, and (c) they are identical fermions.



Problem 5.7 Two noninteracting particles (of equal mass) share the same harmonic oscillator potential, one in the ground state and one in the first excited state.

- (a) Construct the wave function, $\psi(x_1, x_2)$, assuming (i) they are distinguishable, (ii) they are identical bosons, (iii) they are identical fermions. Plot $|\psi(x_1, x_2)|^2$ in each case (use, for instance, Mathematica's **Plot3D**).
 (b) Use Equations 5.23 and 5.25 to determine $\langle (x_1 - x_2)^2 \rangle$ for each case.
 (c) Express each $\psi(x_1, x_2)$ in terms of the relative and center-of-mass coordinates $r \equiv x_1 - x_2$ and $R \equiv (x_1 + x_2)/2$, and integrate over R to get the probability of finding the particles a distance $|r|$ apart:

$$P(|r|) = 2 \int |\psi(R, r)|^2 dR$$

(the 2 accounts for the fact that r could be positive or negative). Graph $P(r)$ for the three cases.

- (d) Define the density operator by

$$n(x) = \sum_{i=1}^2 \delta(x - x_i);$$

$\langle n(x) \rangle dx$ is the expected number of particles in the interval dx . Compute $\langle n(x) \rangle$ for each of the three cases and plot your results. (The result may surprise you.)

⁵ For an incisive critique of this terminology see W. J. Mullin and G. Blaylock, *Am. J. Phys.* **71**, 1223 (2003).

Problem 5.8 Suppose you had *three* particles, one in state $\psi_a(x)$, one in state $\psi_b(x)$, and one in state $\psi_c(x)$. Assuming ψ_a , ψ_b , and ψ_c are orthonormal, construct the three-particle states (analogous to Equations 5.19, 5.20, and 5.21) representing (a) distinguishable particles, (b) identical bosons, and (c) identical fermions. Keep in mind that (b) must be completely symmetric, under interchange of *any* pair of particles, and (c) must be completely *anti*-symmetric, in the same sense. *Comment:* There's a cute trick for constructing completely antisymmetric wave functions: Form the **Slater determinant**, whose first row is $\psi_a(x_1)$, $\psi_b(x_1)$, $\psi_c(x_1)$, etc., whose second row is $\psi_a(x_2)$, $\psi_b(x_2)$, $\psi_c(x_2)$, etc., and so on (this device works for any number of particles).⁶

5.1.3 Spin

It is time to bring spin into the story. The *complete* state of an electron (say) includes not only its position wave function, but also a spinor, describing the orientation of its spin.⁷

$$\psi(\mathbf{r})\chi. \quad (5.27)$$

When we put together the two-particle state,⁸

$$\psi(\mathbf{r}_1, \mathbf{r}_2)\chi(1, 2) \quad (5.28)$$

it is the *whole works*, not just the spatial part, that has to be antisymmetric with respect to exchange:

$$\psi(\mathbf{r}_1, \mathbf{r}_2)\chi(1, 2) = -\psi(\mathbf{r}_2, \mathbf{r}_1)\chi(2, 1). \quad (5.29)$$

Now, a glance back at the composite spin states (Equations 4.175 and 4.176) reveals that the singlet combination is antisymmetric (and hence would have to be joined with a *symmetric* spatial function), whereas the three triplet states are all symmetric (and would require an *anti-symmetric* spatial function). Thus the Pauli principle actually allows *two* electrons in a given position state, as long as their spins are in the singlet configuration (but they could not be in the same position state *and* in the same spin state—say, both spin up).

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Problem 5.9 In Example 5.1 and Problem 5.5(b) we ignored spin (or, if you prefer, we assumed the particles are in the same spin state).

(a) Do it now for particles of spin 1/2. Construct the four lowest-energy configurations, and specify their energies and degeneracies. *Suggestion:* Use the notation $\psi_{n_1 n_2} |s m\rangle$, where $\psi_{n_1 n_2}$ is defined in Example 5.1 and $|s m\rangle$ in Section 4.4.3.⁹

⁶ To construct a completely *symmetric* configuration, use the **permanent** (same as determinant, but without the minus signs).

⁷ In the absence of coupling between spin and position, we are free to assume that the state is *separable* in its spin and spatial coordinates. This just says that the probability of getting spin up is independent of the *location* of the particle. In the *presence* of coupling, the general state would take the form of a linear combination: $\psi_+(\mathbf{r})\chi_+ + \psi_-(\mathbf{r})\chi_-$, as in Problem 4.64.

⁸ I'll let $\chi(1, 2)$ stand for the combined spin state; in Dirac notation it is some linear combination of the states $|s_1 s_2 m_1 m_2\rangle$. I assume that the state is again a simple product of a position state and a spin state; as you'll see in Problem 5.10, this is not always true when three or more electrons are involved—even in the absence of coupling.

⁹ Of course, spin requires three dimensions, whereas we ordinarily think of the infinite square well as existing in one dimension. But it could represent a particle in three dimensions that is confined to a one-dimensional wire.

(b) Do the same for spin 1. *Hint:* First work out the spin-1 analogs to the spin-1/2 singlet and triplet configurations, using the Clebsch–Gordan coefficients; note which of them are symmetric and which antisymmetric.¹⁰

5.1.4 Generalized Symmetrization Principle

I have assumed, for the sake of simplicity, that the particles are noninteracting, the spin and position are decoupled (with the combined state a product of position and spin factors), and the potential is time-independent. But the fundamental symmetrization/antisymmetrization requirement for identical bosons/fermions is much more general. Let us define the **exchange operator**, $\hat{P}$, which interchanges the two particles:¹¹

$$\hat{P} |(1, 2)\rangle = |(2, 1)\rangle. \quad (5.30)$$

Clearly, $\hat{P}^2 = 1$, and it follows (prove it for yourself) that the eigenvalues of $\hat{P}$ are ± 1 . Now, if the two particles are identical, the Hamiltonian must treat them the same: $m_1 = m_2$ and $V(\mathbf{r}_1, \mathbf{r}_2, t) = V(\mathbf{r}_2, \mathbf{r}_1, t)$. It follows that $\hat{P}$ and $\hat{H}$ are compatible observables,

$$[\hat{P}, \hat{H}] = 0, \quad (5.31)$$

and hence (Equation 3.73)

$$\frac{d\langle \hat{P} \rangle}{dt} = 0. \quad (5.32)$$

If the system starts out in an eigenstate of $\hat{P}$ —symmetric ($\langle \hat{P} \rangle = 1$), or antisymmetric ($\langle \hat{P} \rangle = -1$)—then it will stay that way forever. The **symmetrization axiom** says that for identical particles the state is not merely *allowed*, but *required* to satisfy

$$|(1, 2)\rangle = \pm |(2, 1)\rangle, \quad (5.33)$$

with the plus sign for bosons, and the minus sign for fermions.¹² If you have n identical particles, of course, the state must be symmetric or antisymmetric under the interchange of *any two*:

$$|(1, 2, \dots, i, \dots, j, \dots, n)\rangle = \pm |(1, 2, \dots, j, \dots, i, \dots, n)\rangle, \quad (5.34)$$

This is the *general* statement, of which Equation 5.17 is a special case.

¹⁰ This problem was suggested by Greg Elliott.

¹¹ $\hat{P}$ switches the particles ($1 \leftrightarrow 2$); this means exchanging their positions, their spins, and any other properties they might possess. If you like, it switches the *labels*, 1 and 2. I claimed (in Chapter 1) that all our operators would involve multiplication or differentiation; that was a lie. The exchange operator is an exception—and for that matter so is the projection operator (Section 3.6.2).

¹² It is sometimes alleged that the symmetrization requirement (Equation 5.33) is *forced* by the fact that $\hat{P}$ and $\hat{H}$ commute. This is false: It is perfectly possible to imagine a system of two *distinguishable* particles (say, an electron and a positron) for which the Hamiltonian is symmetric, and yet there is no requirement that the state be symmetric (or antisymmetric). But *identical* particles *have* to occupy symmetric or antisymmetric states, and this is a *new fundamental law*—on a par, logically, with Schrödinger's equation and the statistical interpretation. Of course, there didn't *have* to be any such things as identical particles; it could have been that every single particle in the universe was distinguishable from every other one. Quantum mechanics allows for the *possibility* of identical particles, and nature (being lazy) seized the opportunity. (But don't complain—this makes matters enormously simpler!)

**

Problem 5.10 For *two* spin-1/2 particles you can construct symmetric and antisymmetric states (the triplet and singlet combinations, respectively). For *three* spin-1/2 particles you can construct symmetric combinations (the quadruplet, in Problem 4.65), but no completely *anti*-symmetric configuration is possible.

(a) *Prove it. Hint:* The “bulldozer” method is to write down the most general linear combination:

$$\chi(1, 2, 3) = a|\uparrow\uparrow\uparrow\rangle + b|\uparrow\uparrow\downarrow\rangle + c|\uparrow\downarrow\uparrow\rangle + d|\uparrow\downarrow\downarrow\rangle + e|\downarrow\uparrow\uparrow\rangle + f|\downarrow\uparrow\downarrow\rangle + g|\downarrow\downarrow\uparrow\rangle + h|\downarrow\downarrow\downarrow\rangle.$$

What does antisymmetry under $1 \leftrightarrow 2$ tell you about the coefficients? (Note that the eight terms are mutually orthogonal.) Now invoke antisymmetry under $2 \leftrightarrow 3$.

(b) Suppose you put three identical noninteracting spin-1/2 particles in the infinite square well. What is the ground state for this system, what is its energy, and what is its degeneracy? *Note:* You can’t put all three in the position state ψ_1 (why not?); you’ll need two in ψ_1 and the other in ψ_2 . But the symmetric configuration $[\psi_1(x_1)\psi_1(x_2)\psi_2(x_3) + \psi_1(x_1)\psi_2(x_2)\psi_1(x_3) + \psi_2(x_1)\psi_1(x_2)\psi_1(x_3)]$ is no good (because there’s no antisymmetric spin combination to go with it), and you can’t make a completely antisymmetric combination of those three terms. . . . In this case you simply cannot construct an antisymmetric product of a spatial state and a spin state. But you *can* do it with an appropriate *linear combination* of such products. *Hint:* Form the Slater determinant (Problem 5.8) whose top row is $\psi_1(x_1)|\uparrow\rangle_1$, $\psi_1(x_1)|\downarrow\rangle_1$, $\psi_2(x_1)|\uparrow\rangle_1$.

(c) Show that your answer to part (b), properly normalized, can be written in the form

$$\Phi(1, 2, 3) = \frac{1}{\sqrt{3}} [\Phi(1, 2) \phi(3) - \Phi(1, 3) \phi(2) + \Phi(2, 3) \phi(1)]$$

where $\Phi(i, j)$ is the wave function of two particles in the $n = 1$ state and the singlet spin configuration,

$$\Phi(i, j) = \psi_1(x_i) \psi_1(x_j) \frac{|\uparrow_i \downarrow_j\rangle - |\downarrow_i \uparrow_j\rangle}{\sqrt{2}}, \quad (5.35)$$

and $\phi(i)$ is the wave function of the i th particle in the $n = 2$ spin up state: $\phi(i) = \psi_2(x_i) |\uparrow_i\rangle$. Noting that $\Phi(i, j)$ is antisymmetric in $i \leftrightarrow j$, check that $\Phi(1, 2, 3)$ is antisymmetric in all three exchanges ($1 \leftrightarrow 2$, $2 \leftrightarrow 3$, and $3 \leftrightarrow 1$).

**

Problem 5.11 In Section 5.1 we found that for *noninteracting* particles the wave function can be expressed as a product of single-particle states (Equation 5.9)—or, for identical particles, as a symmetrized/antisymmetrized linear combination of such states (Equations 5.20 and 5.21). For *interacting* particles this is no longer the case. A famous example is the **Laughlin wave function**,¹³ which is an approximation to the ground state of N electrons

¹³ “Robert B. Laughlin—Nobel Lecture: Fractional Quantization.” Nobelprize.org. Nobel Media AB 2014. (http://www.nobelprize.org/nobel_prizes/physics/laureates/1998/laughlin-lecture.html).

confined to two dimensions in a perpendicular magnetic field of strength B (the setting for the **fractional quantum Hall effect**). The Laughlin wave function is

$$\psi(z_1, z_2, \dots, z_N) = A \left[\prod_{j < k}^N (z_j - z_k)^q \right] \exp \left[-\frac{1}{2} \sum_k^N |z_k|^2 \right]$$

where q is a positive odd integer and

$$z_j \equiv \sqrt{\frac{eB}{2\hbar c}} (x_j + i y_j) .$$

(Spin is not at issue here; in the ground state all the electrons have spin down with respect to the direction of $\mathbf{B}$, and that is a trivially symmetric configuration.)

- (a) Show that ψ has the proper antisymmetry for fermions.
- (b) For $q = 1$, ψ describes noninteracting particles (by which I mean that it can be written as a single Slater determinant—see Problem 5.8). This is true for any N , but check it explicitly for $N = 3$. What single particle states are occupied in this case?
- (c) For values of q greater than 1, ψ *cannot* be written as a single Slater determinant, and describes interacting particles (in practice, Coulomb repulsion of the electrons). It can, however, be written as a *sum* of Slater determinants. Show that, for $q = 3$ and $N = 2$, ψ can be written as a sum of two Slater determinants.

Comment: In the noninteracting case (b) we can describe the wave function as “three particles occupying the three single-particle states ψ_a , ψ_b and ψ_c ,” but in the interacting case (c), no corresponding statement can be made; in that case, the different Slater determinants that make up ψ correspond to occupation of different sets of single-particle states.

5.2 ATOMS

A neutral atom, of atomic number Z , consists of a heavy nucleus, with electric charge Ze , surrounded by Z electrons (mass m and charge $-e$). The Hamiltonian for this system is¹⁴

$$\hat{H} = \sum_{j=1}^Z \left\{ -\frac{\hbar^2}{2m} \nabla_j^2 - \left(\frac{1}{4\pi\epsilon_0} \right) \frac{Ze^2}{r_j} \right\} + \frac{1}{2} \left(\frac{1}{4\pi\epsilon_0} \right) \sum_{j \neq k}^Z \frac{e^2}{|\mathbf{r}_j - \mathbf{r}_k|} . \quad (5.36)$$

The term in curly brackets represents the kinetic plus potential energy of the j th electron, in the electric field of the nucleus; the second sum (which runs over all values of j and k except $j = k$) is the potential energy associated with the mutual repulsion of the electrons (the factor

¹⁴ I’m assuming the nucleus is *stationary*. The trick of accounting for nuclear motion by using the reduced mass (Problem 5.1) works only for the *two*-body problem; fortunately, the nucleus is so much heavier than the electrons that the correction is extremely small even in the case of hydrogen (see Problem 5.2(a)), and it is smaller still for other atoms. There are more interesting effects, due to magnetic interactions associated with electron spin, relativistic corrections, and the finite size of the nucleus. We’ll look into these in later chapters, but all of them are minute corrections to the “purely coulombic” atom described by Equation 5.36.

of $1/2$ in front corrects for the fact that the summation counts each pair twice). The problem is to solve Schrödinger's equation,

$$\hat{H}\psi = E\psi, \quad (5.37)$$

for the wave function $\psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_Z)$.¹⁵

Unfortunately, the Schrödinger equation with Hamiltonian in Equation 5.36 cannot be solved exactly (at any rate, it *hasn't* been), except for the very simplest case, $Z = 1$ (hydrogen). In practice, one must resort to elaborate approximation methods. Some of these we shall explore in Part II; for now I plan only to sketch some qualitative features of the solutions, obtained by neglecting the electron repulsion term altogether. In Section 5.2.1 we'll study the ground state and excited states of helium, and in Section 5.2.2 we'll examine the ground states of higher atoms.

Problem 5.12

- (a) Suppose you could find a solution $\psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_Z)$ to the Schrödinger equation (Equation 5.37), for the Hamiltonian in Equation 5.36. Describe how you would construct from it a completely symmetric function, and a completely antisymmetric function, which also satisfy the Schrödinger equation, with the same energy. What happens to the completely antisymmetric function if $\psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_Z)$ is symmetric in (say) its first two arguments ($\mathbf{r}_1 \leftrightarrow \mathbf{r}_2$)?
- (b) By the same logic, show that a completely antisymmetric spin state for Z electrons is impossible, if $Z > 2$ (this generalizes Problem 5.10(a)).

5.2.1 Helium

After hydrogen, the simplest atom is helium ($Z = 2$). The Hamiltonian,

$$\hat{H} = \left\{ -\frac{\hbar^2}{2m} \nabla_1^2 - \frac{1}{4\pi\epsilon_0} \frac{2e^2}{r_1} \right\} + \left\{ -\frac{\hbar^2}{2m} \nabla_2^2 - \frac{1}{4\pi\epsilon_0} \frac{2e^2}{r_2} \right\} + \frac{1}{4\pi\epsilon_0} \frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|}, \quad (5.38)$$

consists of two hydrogenic Hamiltonians (with nuclear charge $2e$), one for electron 1 and one for electron 2, together with a final term describing the repulsion of the two electrons. It is this last term that causes all the trouble. If we simply *ignore* it, the Schrödinger equation separates, and the solutions can be written as products of *hydrogen* wave functions:

$$\psi(\mathbf{r}_1, \mathbf{r}_2) = \psi_{n\ell m}(\mathbf{r}_1) \psi_{n'\ell' m'}(\mathbf{r}_2), \quad (5.39)$$

only with half the Bohr radius (Equation 4.72), and four times the Bohr energies (Equation 4.70)—if you don't see why, refer back to Problem 4.19. The total energy would be

$$E = 4(E_n + E_{n'}), \quad (5.40)$$

where $E_n = -13.6/n^2$ eV. In particular, the ground state would be

$$\psi_0(\mathbf{r}_1, \mathbf{r}_2) = \psi_{100}(\mathbf{r}_1) \psi_{100}(\mathbf{r}_2) = \frac{8}{\pi a^3} e^{-2(r_1+r_2)/a}, \quad (5.41)$$

¹⁵ Because the Hamiltonian (5.36) makes no reference to spin, the product $\psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_Z) \chi(s_1, s_2, \dots, s_Z)$ still satisfies the Schrödinger equation. However, for $Z > 2$ such product states cannot in general meet the (anti-)symmetrization requirement, and it is necessary to construct linear combinations, with permuted indices (see Problem 5.16). But that comes at the end of the story; for the moment we are only concerned with the spatial wave function.

(Equation 4.80), and its energy would be

$$E_0 = 8(-13.6 \text{ eV}) = -109 \text{ eV}. \quad (5.42)$$

Because ψ_0 is a symmetric function, the spin state has to be *antisymmetric*, so the ground state of helium should be a *singlet* configuration, with the spins “oppositely aligned.” The *actual* ground state of helium is indeed a singlet, but the experimentally determined energy is -78.975 eV , so the agreement is not very good. But this is hardly surprising: We ignored electron–electron repulsion, which is certainly *not* a small contribution. It is clearly *positive* (see Equation 5.38), which is comforting—evidently it brings the total energy up from -109 to -79 eV (see Problem 5.15).

The excited states of helium consist of one electron in the hydrogenic ground state, and the other in an excited state:

$$\psi_{n\ell m} \psi_{100}. \quad (5.43)$$

(If you try to put *both* electrons in excited states, one immediately drops to the ground state, releasing enough energy to knock the other one into the continuum ($E > 0$), leaving you with a helium *ion* (He^+) and a free electron. This is an interesting system in its own right—see Problem 5.13—but it is not our present concern.) We can construct both symmetric and antisymmetric combinations, in the usual way (Equation 5.17); the former go with the *antisymmetric* spin configuration (the singlet)—they are called **parahelium**—while the latter require a *symmetric* spin configuration (the triplet)—they are known as **orthohelium**. The ground state is necessarily parahelium; the excited states come in both forms. Because the symmetric spatial state brings the electrons closer together (as we discovered in Section 5.1.2), we expect a higher interaction energy in parahelium, and indeed, it is experimentally confirmed that the parahelium states have somewhat higher energy than their orthohelium counterparts (see Figure 5.1).

Problem 5.13

- (a) Suppose you put both electrons in a helium atom into the $n = 2$ state; what would the energy of the emitted electron be? (Assume no photons are emitted in the process.)
- (b) Describe (quantitatively) the spectrum of the helium ion, He^+ . That is, state the “Rydberg-like” formula for the emitted wavelengths.

Problem 5.14 Discuss (qualitatively) the energy level scheme for helium if (a) electrons were identical bosons, and (b) if electrons were distinguishable particles (but with the same mass and charge). Pretend these “electrons” still have spin $1/2$, so the spin configurations are the singlet and the triplet.

**

Problem 5.15

- (a) Calculate $\langle 1/|\mathbf{r}_1 - \mathbf{r}_2| \rangle$ for the state ψ_0 (Equation 5.41). *Hint:* Do the $d^3\mathbf{r}_2$ integral first, using spherical coordinates, and setting the polar axis along $\mathbf{r}_1$, so that

$$|\mathbf{r}_1 - \mathbf{r}_2| = \sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos \theta_2}.$$

The θ_2 integral is easy, but be careful to take the *positive root*. You’ll have to break the r_2 integral into two pieces, one ranging from 0 to r_1 , the other from r_1 to ∞ .
Answer: $5/4a$.

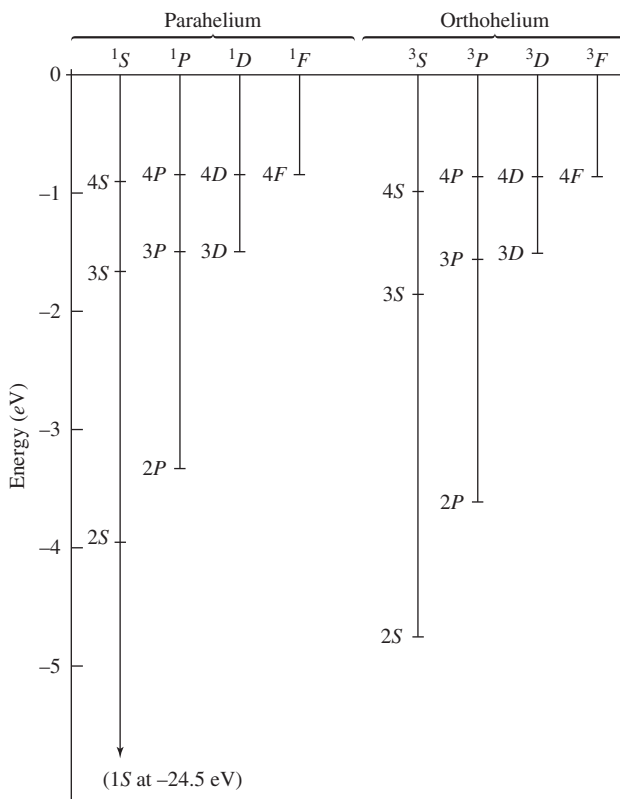


Figure 5.1: Energy level diagram for helium (the notation is explained in Section 5.2.2). Note that parahelium energies are uniformly higher than their orthohelium counterparts. The numerical values on the vertical scale are relative to the ground state of ionized helium (He^+): $4 \times (-13.6) \text{ eV} = -54.4 \text{ eV}$; to get the *total* energy of the state, subtract 54.4 eV.

- (b) Use your result in (a) to estimate the electron interaction energy in the ground state of helium. Express your answer in electron volts, and add it to E_0 (Equation 5.42) to get a corrected estimate of the ground state energy. Compare the experimental value. (Of course, we're still working with an approximate wave function, so don't expect *perfect* agreement.)

Problem 5.16 The ground state of lithium. Ignoring electron–electron repulsion, construct the ground state of lithium ($Z = 3$). Start with a spatial wave function, analogous to Equation 5.41, but remember that only two electrons can occupy the hydrogenic ground state; the third goes to ψ_{200} .¹⁶ What is the energy of this state? Now tack on the spin, and antisymmetrize (if you get stuck, refer back to Problem 5.10). What's the degeneracy of the ground state?

¹⁶ Actually, $\ell = 1$ would do just as well, but electron–electron repulsion favors $\ell = 0$, as we shall see.

5.2.2 The Periodic Table

The ground state electron configurations for heavier atoms can be pieced together in much the same way. To first approximation (ignoring their mutual repulsion altogether) the individual electrons occupy one-particle hydrogenic states (n, ℓ, m) , called **orbitals**, in the Coulomb potential of a nucleus with charge Ze . If electrons were bosons (or distinguishable particles) they would all shake down to the ground state $(1, 0, 0)$, and chemistry would be very dull indeed. But electrons are in fact identical fermions, subject to the Pauli exclusion principle, so only *two* can occupy any given orbital (one with spin up, and one with spin down—or, more precisely, in the singlet configuration). There are n^2 hydrogenic wave functions (all with the same energy E_n) for a given value of n , so the $n = 1$ **shell** has room for two electrons, the $n = 2$ shell holds eight, $n = 3$ takes 18, and in general the n th shell can accommodate $2n^2$ electrons. Qualitatively, the horizontal rows on the **Periodic Table** correspond to filling out each shell (if this were the whole story, they would have lengths 2, 8, 18, 32, 50, etc., instead of 2, 8, 8, 18, 18, etc.; we'll see in a moment how the electron–electron repulsion throws the counting off).

With helium, the $n = 1$ shell is filled, so the next atom, lithium ($Z = 3$), has to put one electron into the $n = 2$ shell. Now, for $n = 2$ we can have $\ell = 0$ or $\ell = 1$; which of these will the third electron choose? In the absence of electron–electron interactions, they have the same energy (the Bohr energies depend on n , remember, but not on ℓ). But the effect of electron repulsion is to favor the lowest value of ℓ , for the following reason. Angular momentum tends to throw the electron outward, and the farther out it gets, the more effectively the inner electrons **screen** the nucleus (roughly speaking, the innermost electron “sees” the full nuclear charge Ze , but the outermost electron sees an effective charge hardly greater than e). Within a given shell, therefore, the state with lowest energy (which is to say, the most tightly bound electron) is $\ell = 0$, and the energy increases with increasing ℓ . Thus the third electron in lithium occupies the orbital $(2, 0, 0)$.¹⁷ The next atom (beryllium, with $Z = 4$) also fits into this state (only with “opposite spin”), but boron ($Z = 5$) has to make use of $\ell = 1$.

Continuing in this way, we reach neon ($Z = 10$), at which point the $n = 2$ shell is filled, and we advance to the next row of the periodic table and begin to populate the $n = 3$ shell. First there are two atoms (sodium and magnesium) with $\ell = 0$, and then there are six with $\ell = 1$ (aluminum through argon). Following argon there “should” be 10 atoms with $n = 3$ and $\ell = 2$; however, by this time the screening effect is so strong that it overlaps the next shell; potassium ($Z = 19$) and calcium ($Z = 20$) choose $n = 4$, $\ell = 0$, in preference to $n = 3$, $\ell = 2$. After that we drop back to pick up the $n = 3$, $\ell = 2$ stragglers (scandium through zinc), followed by $n = 4$, $\ell = 1$ (gallium through krypton), at which point we again make a premature jump to the next row ($n = 5$), and wait until later to slip in the $\ell = 2$ and $\ell = 3$ orbitals from the $n = 4$ shell. For details of this intricate counterpoint I refer you to any book on atomic physics.¹⁸

I would be delinquent if I failed to mention the archaic nomenclature for atomic states, because all chemists and most physicists use it (and the people who make up the Graduate Record Exam *love* this sort of thing). For reasons known best to nineteenth-century spectroscopists, $\ell = 0$ is called *s* (for “sharp”), $\ell = 1$ is *p* (for “principal”), $\ell = 2$ is *d* (“diffuse”), and $\ell = 3$ is *f* (“fundamental”); after that I guess they ran out of imagination, because it now

¹⁷ This standard argument has been called into question by W. Stacey and F. Marsiglio, *EPL*, **100**, 43002 (2012).

¹⁸ See, for example, Ugo Fano and L. Fano, *Basic Physics of Atoms and Molecules*, Wiley, New York (1959), Chapter 18, or the classic by Gerhard Herzberg, *Atomic Spectra and Atomic Structure*, Dover, New York (1944).

continues alphabetically (*g, h, i*, skip *j*, just to be utterly perverse, *k, l*, etc.).¹⁹ The state of a particular electron is represented by the pair $n\ell$, with n (the number) giving the shell, and ℓ (the letter) specifying the orbital angular momentum; the magnetic quantum number m is not listed, but an exponent is used to indicate the number of electrons that occupy the state in question. Thus the configuration

$$(1s)^2 (2s)^2 (2p)^2 \quad (5.44)$$

tells us that there are two electrons in the orbital (1,0,0), two in the orbital (2,0,0), and two in some combination of the orbitals (2,1,1), (2,1,0), and (2,1,−1). This happens to be the ground state of carbon.

In that example there are two electrons with orbital angular momentum quantum number 1, so the *total* orbital angular momentum quantum number, L (capital L —not to be confused with the L denoting $n = 2$ —instead of ℓ , to indicate that this pertains to the *total*, not to any one particle) could be 2, 1, or 0. Meanwhile, the two (1s) electrons are locked together in the singlet state, with total spin zero, and so are the two (2s) electrons, but the two (2p) electrons could be in the singlet configuration or the triplet configuration. So the *total* spin quantum number S (capital, again, because it's the *total*) could be 1 or 0. Evidently the *grand* total (orbital plus spin), J , could be 3, 2, 1, or 0 (Equation 4.182). There exist rituals, known as **Hund's Rules** (see Problem 5.18) for figuring out what these totals will be, for a particular atom. The result is recorded as the following hieroglyphic:

$$^{2S+1}L_J, \quad (5.45)$$

(where S and J are the numbers, and L the letter—capitalized, because we're talking about the *totals*). The ground state of carbon happens to be 3P_0 : the total spin is 1 (hence the 3), the total orbital angular momentum is 1 (hence the P), and the *grand* total angular momentum is zero (hence the 0). In Table 5.1 the individual configurations and the total angular momenta (in the notation of Equation 5.45) are listed, for the first four rows of the Periodic Table.²⁰

*

Problem 5.17

- (a) Figure out the electron configurations (in the notation of Equation 5.44) for the first two rows of the Periodic Table (up to neon), and check your results against Table 5.1.

Table 5.1 *Ground-state electron configurations for the first four rows of the Periodic Table.*

Z	Element	Configuration	
1	H	(1s)	$^2S_{1/2}$
2	He	(1s) ²	1S_0
3	Li	(He)(2s)	$^2S_{1/2}$
4	Be	(He)(2s) ²	1S_0
5	B	(He)(2s) ² (2p)	$^2P_{1/2}$
6	C	(He)(2s) ² (2p) ²	3P_0

¹⁹ The shells themselves are assigned equally arbitrary nicknames, starting (don't ask me why) with K : The K shell is $n = 1$, the L shell is $n = 2$, M is $n = 3$, and so on (at least they're in alphabetical order).

²⁰ After krypton—element 36—the situation gets more complicated (fine structure starts to play a significant role in the ordering of the states) so it is not for want of space that the table terminates there.

7	N	(He)(2s) ² (2p) ³	⁴ S _{3/2}
8	O	(He)(2s) ² (2p) ⁴	³ P ₂
9	F	(He)(2s) ² (2p) ⁵	² P _{3/2}
10	Ne	(He)(2s) ² (2p) ⁶	¹ S ₀
11	Na	(Ne)(3s)	² S _{1/2}
12	Mg	(Ne)(3s) ²	¹ S ₀
13	Al	(Ne)(3s) ² (3p)	² P _{1/2}
14	Si	(Ne)(3s) ² (3p) ²	³ P ₀
15	P	(Ne)(3s) ² (3p) ³	⁴ S _{3/2}
16	S	(Ne)(3s) ² (3p) ⁴	³ P ₂
17	Cl	(Ne)(3s) ² (3p) ⁵	² P _{3/2}
18	Ar	(Ne)(3s) ² (3p) ⁶	¹ S ₀
19	K	(Ar)(4s)	² S _{1/2}
20	Ca	(Ar)(4s) ²	¹ S ₀
21	Sc	(Ar)(4s) ² (3d)	² D _{3/2}
22	Ti	(Ar)(4s) ² (3d) ²	³ F ₂
23	V	(Ar)(4s) ² (3d) ³	⁴ F _{3/2}
24	Cr	(Ar)(4s)(3d) ⁵	⁷ S ₃
25	Mn	(Ar)(4s) ² (3d) ⁵	⁶ S _{5/2}
26	Fe	(Ar)(4s) ² (3d) ⁶	⁵ D ₄
27	Co	(Ar)(4s) ² (3d) ⁷	⁴ F _{9/2}
28	Ni	(Ar)(4s) ² (3d) ⁸	³ F ₄
29	Cu	(Ar)(4s)(3d) ¹⁰	² S _{1/2}
30	Zn	(Ar)(4s) ² (3d) ¹⁰	¹ S ₀
31	Ga	(Ar)(4s) ² (3d) ¹⁰ (4p)	² P _{1/2}
32	Ge	(Ar)(4s) ² (3d) ¹⁰ (4p) ²	³ P ₀
33	As	(Ar)(4s) ² (3d) ¹⁰ (4p) ³	⁴ S _{3/2}
34	Se	(Ar)(4s) ² (3d) ¹⁰ (4p) ⁴	³ P ₂
35	Br	(Ar)(4s) ² (3d) ¹⁰ (4p) ⁵	² P _{3/2}
36	Kr	(Ar)(4s) ² (3d) ¹⁰ (4p) ⁶	¹ S ₀

- (b) Figure out the corresponding total angular momenta, in the notation of Equation 5.45, for the first four elements. List all the *possibilities* for boron, carbon, and nitrogen.

**

Problem 5.18

- (a) **Hund's first rule** says that, consistent with the Pauli principle, the state with the highest total spin (S) will have the lowest energy. What would this predict in the case of the excited states of helium?
- (b) **Hund's second rule** says that, for a given spin, the state with the highest total orbital angular momentum (L), consistent with overall antisymmetrization, will have the lowest energy. Why doesn't carbon have $L = 2$? *Hint*: Note that the "top of the ladder" ($M_L = L$) is symmetric.
- (c) **Hund's third rule** says that if a subshell (n, ℓ) is no more than half filled, then the lowest energy level has $J = |L - S|$; if it is more than half filled, then $J = L + S$ has the lowest energy. Use this to resolve the boron ambiguity in Problem 5.17(b).
- (d) Use Hund's rules, together with the fact that a symmetric spin state must go with an antisymmetric position state (and vice versa) to resolve the carbon and nitrogen ambiguities in Problem 5.17(b). *Hint*: Always go to the "top of the ladder" to figure out the symmetry of a state.

Problem 5.19 The ground state of dysprosium (element 66, in the 6th row of the Periodic Table) is listed as 5I_8 . What are the total spin, total orbital, and grand total angular momentum quantum numbers? Suggest a likely electron configuration for dysprosium.

5.3 SOLIDS

In the solid state, a few of the loosely-bound outermost **valence** electrons in each atom become detached, and roam around throughout the material, no longer subject only to the Coulomb field of a specific “parent” nucleus, but rather to the combined potential of the entire crystal lattice. In this section we will examine two extremely primitive models: first, the “electron gas” theory of Sommerfeld, which ignores *all* forces (except the confining boundaries), treating the wandering electrons as free particles in a box (the three-dimensional analog to an infinite square well); and second, Bloch’s theory, which introduces a periodic potential representing the electrical attraction of the regularly spaced, positively charged, nuclei (but still ignores electron–electron repulsion). These models are no more than the first halting steps toward a quantum theory of solids, but already they reveal the critical role of the Pauli exclusion principle in accounting for “solidity,” and provide illuminating insight into the remarkable electrical properties of conductors, semi-conductors, and insulators.

5.3.1 The Free Electron Gas

Suppose the object in question is a rectangular solid, with dimensions l_x, l_y, l_z , and imagine that an electron inside experiences no forces at all, except at the impenetrable walls:

$$V(x, y, z) = \begin{cases} 0, & 0 < x < l_x, 0 < y < l_y, 0 < z < l_z; \\ \infty, & \text{otherwise.} \end{cases} \quad (5.46)$$

The Schrödinger equation,

$$-\frac{\hbar^2}{2m} \nabla^2 \psi = E \psi,$$

separates, in Cartesian coordinates: $\psi(x, y, z) = X(x)Y(y)Z(z)$, with

$$-\frac{\hbar^2}{2m} \frac{d^2 X}{dx^2} = E_x X, \quad -\frac{\hbar^2}{2m} \frac{d^2 Y}{dy^2} = E_y Y, \quad -\frac{\hbar^2}{2m} \frac{d^2 Z}{dz^2} = E_z Z,$$

and $E = E_x + E_y + E_z$. Letting

$$k_x \equiv \frac{\sqrt{2mE_x}}{\hbar}, \quad k_y \equiv \frac{\sqrt{2mE_y}}{\hbar}, \quad k_z \equiv \frac{\sqrt{2mE_z}}{\hbar},$$

we obtain the general solutions

$$X(x) = A_x \sin(k_x x) + B_x \cos(k_x x), \quad Y(y) = A_y \sin(k_y y) + B_y \cos(k_y y),$$

$$Z(z) = A_z \sin(k_z z) + B_z \cos(k_z z).$$

The boundary conditions require that $X(0) = Y(0) = Z(0) = 0$, so $B_x = B_y = B_z = 0$, and $X(l_x) = Y(l_y) = Z(l_z) = 0$, so

$$k_x l_x = n_x \pi, \quad k_y l_y = n_y \pi, \quad k_z l_z = n_z \pi, \quad (5.47)$$

where each n is a positive integer:

$$n_x = 1, 2, 3, \dots, \quad n_y = 1, 2, 3, \dots, \quad n_z = 1, 2, 3, \dots \quad (5.48)$$

The (normalized) wave functions are

$$\psi_{n_x n_y n_z} = \sqrt{\frac{8}{l_x l_y l_z}} \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right) \sin\left(\frac{n_z \pi}{l_z} z\right), \quad (5.49)$$

and the allowed energies are

$$E_{n_x n_y n_z} = \frac{\hbar^2 \pi^2}{2m} \left(\frac{n_x^2}{l_x^2} + \frac{n_y^2}{l_y^2} + \frac{n_z^2}{l_z^2} \right) = \frac{\hbar^2 k^2}{2m}, \quad (5.50)$$

where k is the magnitude of the **wave vector**, $\mathbf{k} \equiv (k_x, k_y, k_z)$.

If you imagine a three-dimensional space, with axes k_x , k_y , k_z , and planes drawn in at $k_x = (\pi/l_x), (2\pi/l_x), (3\pi/l_x), \dots$, at $k_y = (\pi/l_y), (2\pi/l_y), (3\pi/l_y), \dots$, and at $k_z = (\pi/l_z), (2\pi/l_z), (3\pi/l_z), \dots$, each intersection point represents a distinct (one-particle) stationary state (Figure 5.2). Each block in this grid, and hence also each state, occupies a volume

$$\frac{\pi^3}{l_x l_y l_z} = \frac{\pi^3}{V} \quad (5.51)$$

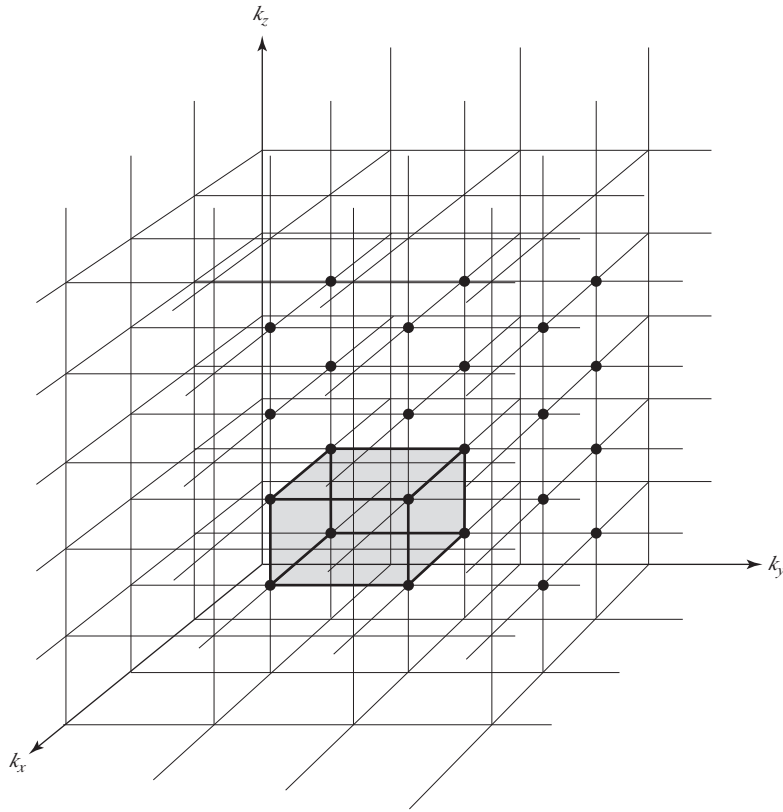


Figure 5.2: Free electron gas. Each intersection on the grid represents a stationary state. The shaded volume is one “block,” and there is one state (potentially two electrons) for every block.

of “ k -space,” where $V \equiv l_x l_y l_z$ is the volume of the object itself. Suppose our sample contains N atoms, and each atom contributes d free electrons. (In practice, N will be enormous—on the order of Avogadro’s number, for an object of macroscopic size—whereas d is a small number—1, 2, or 3, typically.) If electrons were bosons (or distinguishable particles), they would all settle down to the ground state, ψ_{111} .²¹ But electrons are in fact identical *fermions*, subject to the Pauli exclusion principle, so only two of them can occupy any given state. They will fill up one octant of a *sphere* in k -space,²² whose radius, k_F , is determined by the fact that each pair of electrons requires a volume π^3/V (Equation 5.51):

$$\frac{1}{8} \left(\frac{4}{3} \pi k_F^3 \right) = \frac{Nd}{2} \left(\frac{\pi^3}{V} \right).$$

Thus

$$k_F = \left(3\rho\pi^2 \right)^{1/3}, \quad (5.52)$$

where

$$\rho \equiv \frac{Nd}{V} \quad (5.53)$$

is the *free electron density* (the number of free electrons per unit volume).

The boundary separating occupied and unoccupied states, in k -space, is called the **Fermi surface** (hence the subscript F). The corresponding energy is the **Fermi energy**, E_F ; for a free electron gas,

$$E_F = \frac{\hbar^2}{2m} \left(3\rho\pi^2 \right)^{2/3}. \quad (5.54)$$

The *total* energy of the electron gas can be calculated as follows: A shell of thickness dk (Figure 5.3) contains a volume

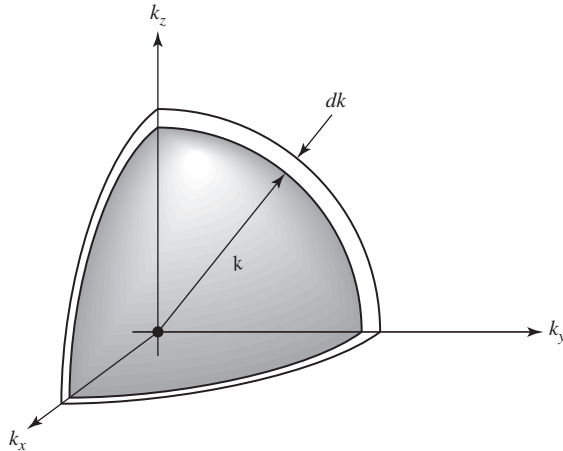


Figure 5.3: One octant of a spherical shell in k -space.

²¹ I’m assuming there is no appreciable thermal excitation, or other disturbance, to lift the solid out of its collective ground state. If you like, I’m talking about a “cold” solid, though (as you will see in Problem 5.21(c)), typical solids are still “cold,” in this sense, far above room temperature.

²² Because N is such a huge number, we need not worry about the distinction between the actual jagged edge of the grid and the smooth spherical surface that approximates it.

$$\frac{1}{8} (4\pi k^2) dk,$$

so the number of electron states in the shell is

$$\frac{2[(1/2)\pi k^2 dk]}{(\pi^3/V)} = \frac{V}{\pi^2} k^2 dk.$$

Each of these states carries an energy $\hbar^2 k^2/2m$ (Equation 5.50), so the energy of the electrons in the shell is

$$dE = \frac{\hbar^2 k^2}{2m} \frac{V}{\pi^2} k^2 dk, \quad (5.55)$$

and hence the total energy of all the filled states is

$$E_{\text{tot}} = \frac{\hbar^2 V}{2\pi^2 m} \int_0^{k_F} k^4 dk = \frac{\hbar^2 k_F^5 V}{10\pi^2 m} = \frac{\hbar^2 (3\pi^2 N d)^{5/3}}{10\pi^2 m} V^{-2/3}. \quad (5.56)$$

This quantum mechanical energy plays a role rather analogous to the internal *thermal* energy (U) of an ordinary gas. In particular, it exerts a *pressure* on the walls, for if the box expands by an amount dV , the total energy decreases:

$$dE_{\text{tot}} = -\frac{2}{3} \frac{\hbar^2 (3\pi^2 N d)^{5/3}}{10\pi^2 m} V^{-5/3} dV = -\frac{2}{3} E_{\text{tot}} \frac{dV}{V},$$

and this shows up as work done on the outside ($dW = P dV$) by the quantum pressure P . Evidently

$$P = \frac{2}{3} \frac{E_{\text{tot}}}{V} = \frac{2}{3} \frac{\hbar^2 k_F^5}{10\pi^2 m} = \frac{(3\pi^2)^{2/3} \hbar^2}{5m} \rho^{5/3}. \quad (5.57)$$

Here, then, is a partial answer to the question of why a cold solid object doesn't simply *collapse*: There is a stabilizing internal pressure, having nothing to do with electron–electron repulsion (which we have ignored) or thermal motion (which we have excluded), but is strictly quantum mechanical, and derives ultimately from the antisymmetrization requirement for the wave functions of identical fermions. It is sometimes called **degeneracy pressure**, though “exclusion pressure” might be a better term.²³

Problem 5.20 Find the average energy per free electron (E_{tot}/Nd), as a fraction of the Fermi energy. *Answer:* $(3/5) E_F$.

Problem 5.21 The density of copper is 8.96 g/cm^3 , and its atomic weight is 63.5 g/mole .

- (a) Calculate the Fermi energy for copper (Equation 5.54). Assume $d = 1$, and give your answer in electron volts.
- (b) What is the corresponding electron velocity? *Hint:* Set $E_F = (1/2) m v^2$. Is it safe to assume the electrons in copper are nonrelativistic?
- (c) At what temperature would the characteristic thermal energy ($k_B T$, where k_B is the Boltzmann constant and T is the Kelvin temperature) equal the Fermi energy, for copper? *Comment:* This is called the **Fermi temperature**, T_F . As long as the *actual* temperature is substantially below the Fermi temperature, the material can be

²³ We *derived* Equations 5.52, 5.54, 5.56, and 5.57 for the special case of an infinite rectangular well, but they hold for containers of any shape, as long as the number of particles is extremely large.

regarded as “cold,” with most of the electrons in the lowest accessible state. Since the melting point of copper is 1356 K, solid copper is *always* cold.

- (d) Calculate the degeneracy pressure (Equation 5.57) of copper, in the electron gas model.

Problem 5.22 Helium-3 is fermion with spin $1/2$ (unlike the more common isotope helium-4 which is a boson). At low temperatures ($T \ll T_F$), helium-3 can be treated as a Fermi gas (Section 5.3.1). Given a density of 82 kg/m^3 , calculate T_F (Problem 5.21(c)) for helium-3.

Problem 5.23 The **bulk modulus** of a substance is the ratio of a small decrease in pressure to the resulting fractional increase in volume:

$$B = -V \frac{dP}{dV}.$$

Show that $B = (5/3)P$, in the free electron gas model, and use your result in Problem 5.21(d) to estimate the bulk modulus of copper. *Comment:* The observed value is $13.4 \times 10^{10} \text{ N/m}^2$, but don’t expect perfect agreement—after all, we’re neglecting all electron–nucleus and electron–electron forces! Actually, it is rather surprising that this calculation comes as close as it *does*.

5.3.2 Band Structure

We’re now going to improve on the free electron model, by including the forces exerted on the electrons by the regularly spaced, positively charged, essentially stationary nuclei. The qualitative behavior of solids is dictated to a remarkable degree by the mere fact that this potential is *periodic*—its actual *shape* is relevant only to the finer details. To show you how it goes, I’m going to develop the simplest possible model: a one-dimensional **Dirac comb**, consisting of evenly spaced delta-function spikes (Figure 5.4).²⁴ But first I need to introduce a powerful theorem that vastly simplifies the analysis of periodic potentials.

A periodic potential is one that repeats itself after some fixed distance a :

$$V(x + a) = V(x). \quad (5.58)$$

Bloch’s theorem tells us that for such a potential the solutions to the Schrödinger equation,

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi = E\psi, \quad (5.59)$$

can be taken to satisfy the condition

$$\psi(x + a) = e^{iqa} \psi(x), \quad (5.60)$$

²⁴ It would be more natural to let the delta functions go *down*, so as to represent the attractive force of the nuclei. But then there would be negative energy solutions as well as positive energy solutions, and that makes the calculations more cumbersome (see Problem 5.26). Since all we’re trying to do here is explore the consequences of periodicity, it is simpler to adopt this less plausible shape; if it comforts you, think of the nuclei as residing at $\pm a/2, \pm 3a/2, \pm 5a/2, \dots$

for some constant q (by “constant” I mean that it is independent of x ; it may well depend on E).²⁵ In a moment we will discover that q is in fact *real*, so although $\psi(x)$ itself is not periodic, $|\psi(x)|^2$ is:

$$|\psi(x+a)|^2 = |\psi(x)|^2, \quad (5.61)$$

as one would certainly expect.²⁶

Of course, no *real* solid goes on forever, and the edges are going to spoil the periodicity of $V(x)$, and render Bloch’s theorem inapplicable. However, for any macroscopic crystal, containing something on the order of Avogadro’s number of atoms, it is hardly imaginable that edge effects can significantly influence the behavior of electrons deep inside. This suggests the following device to salvage Bloch’s theorem: We wrap the x axis around in a circle, and connect it onto its tail, after a large number $N \approx 10^{23}$ of periods; formally, we impose the boundary condition

$$\psi(x + Na) = \psi(x). \quad (5.62)$$

It follows (from Equation 5.60) that

$$e^{iNqa} \psi(x) = \psi(x),$$

so $e^{iNqa} = 1$, or $Nqa = 2\pi n$, and hence

$$q = \frac{2\pi n}{Na}, \quad (n = 0, \pm 1, \pm 2, \dots). \quad (5.63)$$

In particular, q is necessarily real. The virtue of Bloch’s theorem is that we need only solve the Schrödinger equation *within a single cell* (say, on the interval $0 \leq x < a$); recursive application of Equation 5.60 generates the solution everywhere else.

Now, suppose the potential consists of a long string of delta-function spikes (the Dirac comb):

$$V(x) = \alpha \sum_{j=0}^{N-1} \delta(x - ja). \quad (5.64)$$

(In Figure 5.4 you must imagine that the x axis has been “wrapped around”, so the N th spike actually appears at $x = -a$.) No one would pretend that this is a *realistic* model, but remember, it is only the effect of *periodicity* that concerns us here; the classic **Kronig-Penney model**²⁷

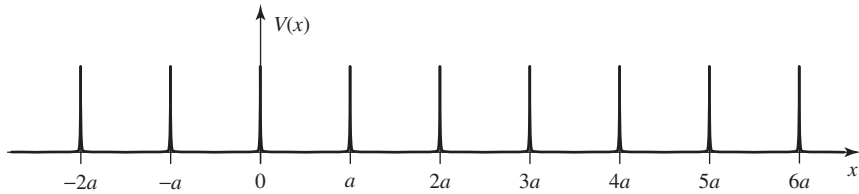


Figure 5.4: The Dirac comb, Equation 5.64.

²⁵ The proof of Bloch’s theorem will come in Chapter 6 (see Section 6.2.2).

²⁶ Indeed, you might be tempted to reverse the argument, *starting* with Equation 5.61, as a way of proving Bloch’s theorem. It doesn’t work, for Equation 5.61 alone would allow the phase factor in Equation 5.60 to be a *function* of x .

²⁷ R. de L. Kronig and W. G. Penney, *Proc. R. Soc. Lond.*, ser. A, **130**, 499 (1930).

used a repeating *rectangular* pattern, and many authors still prefer that one.²⁸ In the region $0 < x < a$ the potential is zero, so

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi,$$

or

$$\frac{d^2\psi}{dx^2} = -k^2\psi,$$

where

$$k \equiv \frac{\sqrt{2mE}}{\hbar}, \quad (5.65)$$

as usual.

The general solution is

$$\psi(x) = A \sin(kx) + B \cos(kx), \quad (0 < x < a). \quad (5.66)$$

According to Bloch's theorem, the wave function in the cell immediately to the *left* of the origin is

$$\psi(x) = e^{-iqa} \{A \sin[k(x+a)] + B \cos[k(x+a)]\}, \quad (-a < x < 0). \quad (5.67)$$

At $x = 0$, ψ must be continuous, so

$$B = e^{-iqa} [A \sin(ka) + B \cos(ka)]; \quad (5.68)$$

its derivative suffers a discontinuity proportional to the strength of the delta function (Equation 2.128, with the sign of α switched, since these are spikes instead of wells):

$$kA - e^{-iqa}k[A \cos(ka) - B \sin(ka)] = \frac{2m\alpha}{\hbar^2}B. \quad (5.69)$$

Solving Equation 5.68 for $A \sin(ka)$ yields

$$A \sin(ka) = [e^{iqa} - \cos(ka)]B. \quad (5.70)$$

Substituting this into Equation 5.69, and cancelling kB , we find

$$[e^{iqa} - \cos(ka)] [1 - e^{-iqa} \cos(ka)] + e^{-iqa} \sin^2(ka) = \frac{2m\alpha}{\hbar^2 k} \sin(ka),$$

which simplifies to

$$\cos(qa) = \cos(ka) + \frac{m\alpha}{\hbar^2 k} \sin(ka). \quad (5.71)$$

This is the fundamental result, from which all else follows.²⁹

Equation 5.71 determines the possible values of k , and hence the allowed energies. To simplify the notation, let

$$z \equiv ka, \quad \text{and} \quad \beta \equiv \frac{m\alpha a}{\hbar^2}, \quad (5.72)$$

so the right side of Equation 5.71 can be written as

$$f(z) \equiv \cos(z) + \beta \frac{\sin(z)}{z}. \quad (5.73)$$

²⁸ See, for instance, David Park, *Introduction to the Quantum Theory*, 3rd edn, McGraw-Hill, New York (1992).

²⁹ For the Kronig–Penney potential (footnote 27, page 221), the formula is more complicated, but it shares the qualitative features we are about to explore.

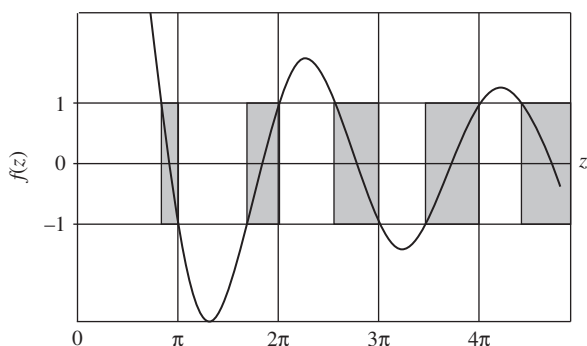


Figure 5.5: Graph of $f(z)$ (Equation 5.73) for $\beta = 10$, showing allowed bands separated by forbidden gaps.

The constant β is a dimensionless measure of the “strength” of the delta function. In Figure 5.5 I have plotted $f(z)$, for the case $\beta = 10$. The important thing to notice is that $f(z)$ strays outside the range $(-1, +1)$, and in such regions there is no hope of solving Equation 5.71, since $|\cos(qa)|$, of course, cannot be greater than 1. These **gaps** represent *forbidden* energies; they are separated by **bands** of *allowed* energies. Within a given band, virtually any energy is allowed, since according to Equation 5.63, $qa = 2\pi n/N$, where N is a huge number, and n can be any integer. You might imagine drawing N horizontal lines on Figure 5.5, at values of $\cos(2\pi n/N)$ ranging from $+1$ ($n = 0$) down to -1 ($n = N/2$), and back almost to $+1$ ($n = N - 1$)—at this point the Bloch factor e^{iqa} recycles, so no new solutions are generated by further increasing n . The intersection of each of these lines with $f(z)$ yields an allowed energy. Evidently there are N states in each band, so closely spaced that for most purposes we can regard them as forming a continuum (Figure 5.6).

So far, we’ve only put *one* electron in our potential. In practice there will be Nd of them, where d is again the number of “free” electrons per atom. Because of the Pauli exclusion principle, only two electrons can occupy a given spatial state, so if $d = 1$, they will half fill the first band, if $d = 2$ they will completely fill the first band, if $d = 3$ they half fill the second band, and so on. (In three dimensions, and with more realistic potentials, the band structure may be more complicated, but the *existence* of allowed bands, separated by forbidden gaps, persists—band structure is the *signature* of a periodic potential.³⁰)

Now, if the topmost band is only *partly* filled, it takes very little energy to excite an electron to the next allowed level, and such a material will be a **conductor** (a **metal**). On the other hand, if the top band is *completely* filled, it takes a relatively large energy to excite an electron, since it has to jump across the forbidden zone. Such materials are typically **insulators**, though if the gap is rather narrow, and the temperature sufficiently high, then random thermal energy can knock an electron over the hump, and the material is a **semiconductor** (silicon and germanium are examples).³¹ In the free electron model *all* solids should be metals, since there are no

³⁰ Regardless of dimension, if d is an odd integer you are guaranteed to have partially-filled bands and you would expect metallic behavior. If d is an even integer, it depends on the specific band structure whether there will be partially-filled bands or not. Interestingly, some materials, called **Mott insulators**, are nonconductors even though d is odd. In that case it is the interactions between electrons that leads to the insulating behavior, not the presence of gaps in the single-particle energy spectrum.

³¹ Semiconductors typically have band gaps of 4 eV or less, small enough that thermal excitation at room temperature ($k_B T \approx 0.025$ eV) produces perceptible conductivity. The conductivity of a semiconductor can be controlled by

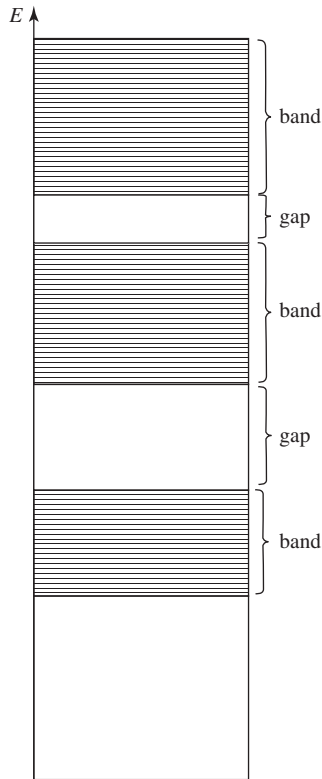


Figure 5.6: The allowed energies for a periodic potential form essentially continuous bands.

large gaps in the spectrum of allowed energies. It takes the band theory to account for the extraordinary range of electrical conductivities exhibited by the solids in nature.

Problem 5.24

- (a) Using Equations 5.66 and 5.70, show that the wave function for a particle in the periodic delta function potential can be written in the form

$$\psi(x) = C \left\{ \sin(kx) + e^{-iqa} \sin[k(a-x)] \right\}, \quad (0 \leq x \leq a).$$

(Don't bother to determine the normalization constant C .)

- (b) At the top of a band, where $z = j\pi$, (a) yields $\psi(x) = 0/0$ (indeterminate). Find the correct wave function for this case. Note what happens to ψ at each delta function.

Problem 5.25 Find the energy at the bottom of the first allowed band, for the case $\beta = 10$, correct to three significant digits. For the sake of argument, assume $\alpha/a = 1$ eV.

doping: including a few atoms of larger or smaller d ; this puts some “extra” electrons into the next higher band, or creates some **holes** in the previously filled one, allowing in either case for weak electric currents to flow.

**

Problem 5.26 Suppose we use delta function *wells*, instead of *spikes* (i.e. switch the sign of α in Equation 5.64). Analyze this case, constructing the analog to Figure 5.5. This requires no new calculation, for the positive energy solutions (except that β is now negative; use $\beta = -1.5$ for the graph), but you *do* need to work out the negative energy solutions (let $\kappa \equiv \sqrt{-2mE}/\hbar$ and $z \equiv -\kappa a$, for $E < 0$); your graph will now extend to negative z). How many states are there in the first allowed band?

Problem 5.27 Show that *most* of the energies determined by Equation 5.71 are doubly degenerate. What are the exceptional cases? *Hint:* Try it for $N = 1, 2, 3, 4, \dots$, to see how it goes. What are the possible values of $\cos(qa)$ in each case?



Problem 5.28 Make a plot of E vs. q for the band structure in Section 5.3.2. Use $\alpha = 1$ (in units where $m = \hbar = a = 1$). *Hint:* In Mathematica, **ContourPlot** will graph $E(q)$ as defined implicitly by Equation 5.71. On other platforms the plot can be obtained as follows:

- Choose a large number (say 30,000) of equally-spaced values for the energy in the range $E = 0$ and $E = 30$.
- For each value of E , compute the right-hand side of Equation 5.71. If the result is between -1 and 1 , solve for q from Equation 5.71 and record the pair of values $\{q, E\}$ and $\{-q, E\}$ (there are two solutions for each energy).

You will then have a list of pairs $\{\{q_1, E_1\}, \{q_2, E_2\}, \dots\}$ which you can plot.

FURTHER PROBLEMS ON CHAPTER 5

Problem 5.29 Suppose you have three particles, and three distinct one-particle states ($\psi_a(x)$, $\psi_b(x)$, and $\psi_c(x)$) are available. How many different three-particle states can be constructed (a) if they are distinguishable particles, (b) if they are identical bosons, (c) if they are identical fermions? (The particles need not be in *different* states— $\psi_a(x_1)\psi_a(x_2)\psi_a(x_3)$ would be one possibility, if the particles are distinguishable.)

Problem 5.30 Calculate the Fermi energy for electrons in a *two*-dimensional infinite square well. Let σ be the number of free electrons per unit area.

Problem 5.31 Repeat the analysis of Problem 2.58 to estimate the cohesive energy for a three-dimensional metal, including the effects of spin.

Problem 5.32 Consider a free electron gas (Section 5.3.1) with unequal numbers of spin-up and spin-down particles (N_+ and N_- respectively). Such a gas would have a net **magnetization** (magnetic dipole moment per unit volume)

$$\mathbf{M} = - \frac{(N_+ - N_-)}{V} \mu_B \hat{k} = M \hat{k}, \quad (5.74)$$

where $\mu_B = e \hbar / 2 m_e$ is the **Bohr magneton**. (The minus sign is there, of course, because the charge of the electron is negative.)

- (a) Assuming that the electrons occupy the lowest energy levels consistent with the number of particles in each spin orientation, find E_{tot} . Check that your answer reduces to Equation 5.56 when $N_+ = N_-$.
- (b) Show that for $M/\mu_B \ll \rho \equiv (N_+ + N_-)/V$ (which is to say, $|N_+ - N_-| \ll (N_+ + N_-)$), the energy density is

$$\frac{1}{V} E_{\text{tot}} = \frac{\hbar^2 (3 \pi^2 \rho)^{5/3}}{10 \pi^2 m} \left[1 + \frac{5}{9} \left(\frac{M}{\rho \mu_B} \right)^2 \right].$$

The energy is a minimum for $M = 0$, so the ground state will have zero magnetization. However, if the gas is placed in a magnetic field (or in the presence of interactions between the particles) it may be energetically favorable for the gas to magnetize. This is explored in Problems 5.33 and 5.34.

Problem 5.33 Pauli paramagnetism. If the free electron gas (Section 5.3.1) is placed in a uniform magnetic field $\mathbf{B} = B \hat{k}$, the energies of the spin-up and spin-down states will be different:³²

$$E_{n_x n_y n_z}^{\pm} = \frac{\hbar^2 \pi^2}{2m} \left(\frac{n_x^2}{l_x^2} + \frac{n_y^2}{l_y^2} + \frac{n_z^2}{l_z^2} \right) \pm \mu_B B.$$

There will be more spin-down states occupied than spin-up states (since they are lower in energy), and consequently the system will acquire a magnetization (see Problem 5.32).

- (a) In the approximation that $M/\mu_B \ll \rho$, find the magnetization that minimizes the total energy. *Hint:* Use the result of Problem 5.32(b).
- (b) The **magnetic susceptibility** is³³

$$\chi = \mu_0 \frac{dM}{dB}.$$

Calculate the magnetic susceptibility for aluminum ($\rho = 18.1 \times 10^{22} \text{ cm}^{-3}$) and compare the experimental value³⁴ of 22×10^{-6} .

³² Here we are considering only the coupling of the spin to the magnetic field, and ignoring any coupling of the orbital motion.

³³ Strictly speaking, the susceptibility is dM/dH , but the difference is negligible when, as here, $\chi \ll 1$.

³⁴ For some metals, such as copper, the agreement is not so good—even the sign is wrong: copper is diamagnetic ($\chi < 0$). The explanation for this discrepancy lies in what has been left out of our model. In addition to the

Problem 5.34 The Stoner criterion. The free-electron gas model (Section 5.3.1) ignores the Coulomb repulsion between electrons. Because of the exchange force (Section 5.1.2), Coulomb repulsion has a stronger effect on two electrons with antiparallel spins (which behave in a way like distinguishable particles) than two electrons with parallel spins (whose position wave function must be antisymmetric). As a crude way to take account of Coulomb repulsion, pretend that every pair of electrons with opposite spin carries extra energy U , while electrons with the same spin do not interact at all; this adds $\Delta E = U N_+ N_-$ to the total energy of the electron gas. As you will show, above a critical value of U , it becomes energetically favorable for the gas to spontaneously magnetize ($N_+ \neq N_-$); the material becomes ferromagnetic.

- (a) Rewrite ΔE in terms of the density ρ and the magnetization M (Equation 5.74).
- (b) Assuming that $M/\mu_B \ll \rho$, for what minimum value of U is a non-zero magnetization energetically favored? *Hint:* Use the result of Problem 5.32(b).

Problem 5.35 Certain cold stars (called **white dwarfs**) are stabilized against gravitational collapse by the degeneracy pressure of their electrons (Equation 5.57). Assuming constant density, the radius R of such an object can be calculated as follows:

- (a) Write the total electron energy (Equation 5.56) in terms of the radius, the number of nucleons (protons and neutrons) N , the number of electrons per nucleon d , and the mass of the electron m . *Beware:* In this problem we are recycling the letters N and d for a slightly different purpose than in the text.
- (b) Look up, or calculate, the gravitational energy of a uniformly dense sphere. Express your answer in terms of G (the constant of universal gravitation), R , N , and M (the mass of a nucleon). Note that the gravitational energy is *negative*.
- (c) Find the radius for which the total energy, (a) plus (b), is a minimum. *Answer:*

$$R = \left(\frac{9\pi}{4} \right)^{2/3} \frac{\hbar^2 d^{5/3}}{GmM^2 N^{1/3}}.$$

(Note that the radius *decreases* as the total mass *increases*!) Put in the actual numbers, for everything except N , using $d = 1/2$ (actually, d decreases a bit as the atomic number increases, but this is close enough for our purposes). *Answer:* $R = 7.6 \times 10^{25} N^{-1/3}$ m.

- (d) Determine the radius, in kilometers, of a white dwarf with the mass of the sun.
- (e) Determine the Fermi energy, in electron volts, for the white dwarf in (d), and compare it with the rest energy of an electron. Note that this system is getting dangerously relativistic (see Problem 5.36).

paramagnetic coupling of the spin magnetic moment to an applied field there is a coupling of the orbital magnetic moment to an applied field and this has both paramagnetic and diamagnetic contributions (see Problem 4.72). In addition, the free electron gas model ignores the tightly-bound core electrons and these also couple to the magnetic field. In the case of copper, it is the diamagnetic coupling of the core electrons that dominates.

Problem 5.36 We can extend the theory of a free electron gas (Section 5.3.1) to the relativistic domain by replacing the classical kinetic energy, $E = p^2/2m$, with the relativistic formula, $E = \sqrt{p^2c^2 + m^2c^4} - mc^2$. Momentum is related to the wave vector in the usual way: $\mathbf{p} = \hbar\mathbf{k}$. In particular, in the *extreme* relativistic limit, $E \approx pc = \hbar ck$.

- (a) Replace $\hbar^2k^2/2m$ in Equation 5.55 by the ultra-relativistic expression, $\hbar ck$, and calculate E_{tot} in this regime.
- (b) Repeat parts (a) and (b) of Problem 5.35 for the ultra-relativistic electron gas. Notice that in this case there is *no* stable minimum, regardless of R ; if the total energy is positive, degeneracy forces exceed gravitational forces, and the star will expand, whereas if the total is negative, gravitational forces win out, and the star will collapse. Find the critical number of nucleons, N_c , such that gravitational collapse occurs for $N > N_c$. This is called the **Chandrasekhar limit**. *Answer:* 2.04×10^{57} . What is the corresponding stellar mass (give your answer as a multiple of the sun's mass). Stars heavier than this will not form white dwarfs, but collapse further, becoming (if conditions are right) **neutron stars**.
- (c) At extremely high density, **inverse beta decay**, $e^- + p^+ \rightarrow n + \nu$, converts virtually all of the protons and electrons into neutrons (liberating neutrinos, which carry off energy, in the process). Eventually *neutron* degeneracy pressure stabilizes the collapse, just as *electron* degeneracy does for the white dwarf (see Problem 5.35). Calculate the radius of a neutron star with the mass of the sun. Also calculate the (neutron) Fermi energy, and compare it to the rest energy of a neutron. Is it reasonable to treat a neutron star nonrelativistically?



Problem 5.37 An important quantity in many calculations is the **density of states** $G(E)$:

$$G(E) dE \equiv \text{number of states with energy between } E \text{ and } E + dE.$$

For a one-dimensional band structure,

$$G(E) dE = 2 \left(\frac{dq}{2\pi/N a} \right),$$

where $dq/(2\pi/N a)$ counts the number of states in the range dq (see Equation 5.63), and the factor of 2 accounts for the fact that states with q and $-q$ have the same energy. Therefore

$$\frac{1}{N a} G(E) = \frac{1}{\pi} \frac{1}{|dE/dq|}.$$

- (a) Show that for $\alpha = 0$ (a free particle) the density of states is given by

$$\frac{1}{N a} G_{\text{free}}(E) = \frac{1}{\pi \hbar} \sqrt{\frac{m}{2E}}.$$

- (b) Find the density of states for $\alpha \neq 0$ by differentiating Equation 5.71 with respect to q to determine dE/dq . *Note:* Your answer should be written as a function of E only (well, and α , m , $\hbar$, a , and N) and must not contain q (use k as a shorthand for $\sqrt{2mE}/\hbar$, if you like).

- (c) Make a single plot showing $G(E)/Na$ for both $\alpha = 0$ and $\alpha = 1$ (in units where $m = \hbar = a = 1$). *Comment:* The divergences at the band edges are examples of **van Hove singularities**.³⁵

Problem 5.38 The **harmonic chain** consists of N equal masses arranged along a line and connected to their neighbors by identical springs:

$$\hat{H} = -\frac{\hbar^2}{2m} \sum_{j=1}^N \frac{\partial}{\partial x_j^2} + \sum_{j=1}^N \frac{1}{2} m \omega^2 (x_{j+1} - x_j)^2,$$

where x_j is the displacement of the j th mass from its equilibrium position. This system (and its extension to two or three dimensions—the **harmonic crystal**) can be used to model the vibrations of a solid. For simplicity we will use periodic boundary conditions: $x_{N+1} = x_1$, and introduce the ladder operators³⁶

$$\hat{a}_{k\pm} \equiv \frac{1}{\sqrt{N}} \sum_{j=1}^N e^{\pm i 2\pi j k/N} \left[\sqrt{\frac{m \omega_k}{2 \hbar}} x_j \mp \sqrt{\frac{\hbar}{2 m \omega_k}} \frac{\partial}{\partial x_j} \right] \quad (5.75)$$

where $k = 1, \dots, N-1$ and the frequencies are given by

$$\omega_k = 2 \omega \sin \left(\frac{\pi k}{N} \right).$$

- (a) Prove that, for integers k and k' between 1 and $N-1$,

$$\begin{aligned} \frac{1}{N} \sum_{j=1}^N e^{i 2\pi j (k-k')/N} &= \delta_{k',k} \\ \frac{1}{N} \sum_{j=1}^N e^{i 2\pi j (k+k')/N} &= \delta_{k',N-k}. \end{aligned}$$

Hint: Sum the geometric series.

- (b) Derive the commutation relations for the ladder operators:

$$[\hat{a}_{k-}, \hat{a}_{k'+}] = \delta_{k,k'} \quad \text{and} \quad [\hat{a}_{k-}, \hat{a}_{k'-}] = [\hat{a}_{k+}, \hat{a}_{k'+}] = 0. \quad (5.76)$$

³⁵ These one-dimensional Van Hove singularities have been observed in the spectroscopy of carbon nanotubes; see J. W. G. Wildöer et al., *Nature*, **391**, 59 (1998).

³⁶ If you are familiar with the *classical* problem of coupled oscillators, these ladder operators are straightforward to construct. Start with the normal mode coordinates you would use to decouple the classical problem, namely

$$q_k = \frac{1}{\sqrt{N}} \sum_{j=1}^N e^{-i 2\pi j k/N} x_j.$$

The frequencies ω_k are the classical normal mode frequencies, and you simply create a pair of ladder operators for each normal mode, by analogy with the single-particle case (Equation 2.48).

(c) Using Equation 5.75, show that

$$x_j = R + \frac{1}{\sqrt{N}} \sum_{k=1}^{N-1} \sqrt{\frac{\hbar}{2m\omega_k}} (\hat{a}_{k-} + \hat{a}_{N-k+}) e^{i2\pi jk/N}$$

$$\frac{\partial}{\partial x_j} = \frac{1}{N} \frac{\partial}{\partial R} + \frac{1}{\sqrt{N}} \sum_{k=1}^{N-1} \sqrt{\frac{m\omega_k}{2\hbar}} (\hat{a}_{k-} - \hat{a}_{N-k+}) e^{i2\pi jk/N}$$

where $R = \sum_j x_j / N$ is the center of mass coordinate.

(d) Finally, show that

$$\hat{H} = -\frac{\hbar^2}{2(Nm)} \frac{\partial^2}{\partial R^2} + \sum_{k=1}^{N-1} \hbar\omega_k \left(\hat{a}_{k+} \hat{a}_{k-} + \frac{1}{2} \right).$$

Comment: Written in this form above, the Hamiltonian describes $N - 1$ independent oscillators with frequencies ω_k (as well as a center of mass that moves as a free particle of mass Nm). We can immediately write down the allowed energies:

$$E = -\frac{\hbar^2 K^2}{2(Nm)} + \sum_{k=1}^{N-1} \hbar\omega_k \left(n_k + \frac{1}{2} \right)$$

where $\hbar K$ is the momentum of the center of mass and $n_k = 0, 1, \dots$ is the energy level of the k th mode of vibration. It is conventional to call n_k the **number of phonons** in the k th mode. Phonons are the quanta of sound (atomic vibrations), just as photons are the quanta of light. The ladder operators a_{k+} and a_{k-} are called **phonon creation and annihilation operators** since they increase or decrease the number of phonons in the k th mode.

Problem 5.39 In Section 5.3.1 we put the electrons in a box with impenetrable walls. The same results can be obtained using **periodic boundary conditions**. We still imagine the electrons to be confined to a box with sides of length l_x , l_y , and l_z but instead of requiring the wave function to vanish on each wall, we require it to take the same value on opposite walls:

$$\psi(x, y, z) = \psi(x + l_x, y, z) = \psi(x, y + l_y, z) = \psi(x, y, z + l_z).$$

In this case we can represent the wave functions as *traveling* waves,

$$\psi = \frac{1}{\sqrt{l_x l_y l_z}} e^{i\mathbf{k} \cdot \mathbf{r}} = \frac{1}{\sqrt{l_x l_y l_z}} e^{i(k_x x + k_y y + k_z z)},$$

rather than as *standing* waves (Equation 5.49). Periodic boundary conditions—while certainly not physical—are often easier to work with (to describe something like electrical current a basis of traveling waves is more natural than a basis of standing waves) and if you are computing *bulk* properties of a material it shouldn't matter which you use.

- (a)** Show that with periodic boundary conditions the wave vector satisfies

$$k_x l_x = 2 n_x \pi, \quad k_y l_y = 2 n_y \pi, \quad k_z l_z = 2 n_z \pi,$$

where each n is an integer (*not* necessarily positive). What is the k -space volume occupied by each block on the grid (corresponding to Equation 5.51)?

- (b)** Compute k_F , E_F , and E_{tot} for the free electron gas with periodic boundary conditions. What compensates for the larger volume occupied by each k -space block (part (a)) to make these all come out the same as in Section 5.3.1?